Woo Kim

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SUMMARY

Machine Learning Engineer with 10+ years of experience building end-to-end ML systems and full-stack applications, enabling data-driven experimentation, model training, and scalable evaluation systems. Experienced in translating research into production-ready tools used by cross-functional teams.

PROFESSIONAL EXPERIENCE

ML Research Engineer (Contract)| **Meta** | Redmond, WA | Dec 2023 – Dec 2025

- Built end-to-end ML workflows for hand pose estimation, including data curation, training, evaluation, and iterative model improvement, enabling rapid experimentation and reliable deployment.
- Improved model performance by refining datasets and optimizing training pipelines, reducing regression error by >50% and increasing robustness under severe occlusions.
- Engineered multimodal data pipelines integrating mocap, tactile, IMU, and optical fiber data into time-synchronized streams, supporting consistent data collection and evaluation workflows.
- Accelerated experimentation by migrating workflows to multi-GPU infrastructure and building tools for benchmarking, experiment tracking, and performance analysis.
- Developed automated annotation systems to replace manual labeling, significantly reducing data generation costs while maintaining high-quality datasets.
- Contributed to team scalability by onboarding and mentoring engineers and researchers, and by documenting ML workflows, tools, and infrastructure.

Software Engineer II (FTE)| **Microsoft** | Redmond, WA | Jan 2018 – Apr 2023

- Led early engineering efforts for Science Engine, building a full-stack enterprise AI platform from prototype to production, driving adoption across global enterprise clients.
- Designed and implemented end-to-end product features spanning React/TypeScript frontend and ASP.NET backend services, allowing users to launch workflows, visualize results, and interact with ML-driven systems.
- Designed core backend systems, including lifecycle management, telemetry pipelines, and error propagation, ensuring reliability and observability in production environments.
- Built telemetry dashboards and monitoring tools for system workflows, enabling managers and engineers to track activity, prioritize issues, and debug system behavior with real-time visibility.
- Developed and maintained high-performance ASP.NET microservices with distributed cache and cloud storage on Azure, supporting scalable and reliable system operation.

Senior Software Engineer | **Huawei** | Bellevue, WA | Sep 2017 – Nov 2017

- Implemented facial alignment models using Caffe and PyTorch, translating research algorithms into working prototypes.
- Optimized model performance for edge deployment, reducing inference latency by 3% without accuracy loss through targeted architectural adjustments.

Software Design Engineer (Contract) | **Microsoft** | Bellevue, WA | Apr 2016 – Jul 2017

- Developed real-time computer vision prototypes (scene understanding, object detection) in close collaboration with research teams, supporting interactive demos and rapid experimentation.
- Built an end-to-end indoor positioning and sensing system integrating UWB sensors, embedded hardware (Arduino), backend services, and cross-platform UX (Qt, iOS, HoloLens).
- Developed ML evaluation and data annotation tools with intuitive UX, improving research workflow efficiency and iteration speed.
- Led and trained a 7-member team to deliver a high-quality annotated dataset under tight deadlines.

Computer Vision Software Engineer | **SRI International** | Princeton, NJ | Jul 2014 – Oct 2014
▪ Developed an end-to-end ML pipeline for generating 3D facial expressions from 2D video, covering data alignment, 3D reconstruction, model training, and a Unity-based user application.
▪ Independently led all R&D efforts across Python, C++, and graphics tools (Blender, Unity) in a remote environment.

Research Scientist | **KIST** | Seoul, South Korea | Dec 2012 – Dec 2013
▪ Developed a novel hand pose estimation algorithm using random forest on point cloud data from Kinect and RealSense cameras.
▪ Generated synthetic hand gesture datasets by simulating RGB-D sensor signals and articulated 3D hand models, compressed using octrees for improved data efficiency.

EDUCATION

MBA (Evening Program) | **University of Washington** | Seattle, WA | May 2019
▪ Dean's List student (top 10% GPA)
▪ Vice president of Foster Tech Club (organized tech events for 100+ students)

M.S. in Biometrics | **Yonsei University** | Seoul, Korea | Feb 2012
▪ Thesis: "3D Head Pose Estimation Using Adaptive Face Template"
▪ Awards: Brain Korea 21 scholarship
▪ Corporate projects:
  - Head pose estimation for Samsung Galaxy SIII with Samsung Electronics
  - Unscented Kalman Filter for object tracking with LG Electronics
▪ Publications:
  - "Automatic head pose estimation from a single camera using projective geometry," *2011 8th International Conference on Information, Communications & Signal Processing*, Singapore
  - "Robust 3D face shape estimation using multiple deformable models," *2011 6th IEEE Conference on Industrial Electronics and Applications*, Beijing, China

B.S. in Computer Engineering | **University of Illinois at Urbana-Champaign** | Urbana, IL | Dec 2008

TECHNICAL SKILLS

Machine Learning & Perception
ML pipelines, data pipelines, model evaluation, distributed training, CNN, hand pose estimation, gesture recognition, facial alignment network, scene understanding (C3D), object detection (YOLO), point-cloud object detection, Kalman filter tracking, head pose estimation (POSIT), epipolar geometry, RBF interpolation

Programming
C++, Python, C#, TypeScript, JavaScript, ASP.NET, REST APIs, Cosmos DB, Redis, Azure Blob Storage, React, React Native, Android Studio

Frameworks & Tools
PyTorch, NumPy, OpenCV, Caffe, Azure, Qt, Unity, Blender, OpenGL, R, MATLAB